

SANTHOSHINI PRANATHI

singarajupranathi@gmail.com ·  [LinkedIn](#) ·  [Github](#)

EDUCATION

Rajiv Gandhi University Of Knowledge and Technologies

Aug 2022 – Present

Andhra Pradesh, India

- Bachelor of Technology in **Electronics and Communication** and more fascinated in CS.
- CGPA: 8.3

Pre University Course

Jun 2020 – Jun 2022

GPA: 9.57

SKILL SUMMARY

- **Languages** Python, C++, C, CUDA programming, EDA
- **Frameworks** PyTorch, CUDA, Triton, CUTLASS, TensorRT-LLM, cuBLAS
- **Open Libraries** NumPy, Pandas, Matplotlib
- **Software** Linux, GitHub
- **Communication** Technical writing, research presentation, and efficient fluency of english

INTERNSHIPS

QML based Behavioral Analytics for Insider-Threat Detection in Cloud Storage

Sep 2025 – Present

Research Intern at BITS Pilani

- Authored- **MAD-EBA**, a detector-agnostic behavioral anomaly detection framework for smart cloud systems, improving anomaly detection F1-score by up to 9.5% across Isolation Forest and Autoencoder-based models on 2,425 entity-period profiles; accepted for poster publication.
- Developed **FedPAE**, a privacy-preserving federated personalized autoencoder framework for distributed cloud systems, achieving 5–13% higher anomaly detection performance over FedAvg/FedProx baselines on CMU CERT and cloud storage datasets; accepted and published at ICTIS 2026 conference. [Papers](#).

Unified Mentor Internship

Dec 2024 – Feb 2025

- Worked on Super Market Grocery Sales, drugs and instagram influencers datasets and data analytics.
- achieved 1.35x more model performance and accuracy. [Letter Of recommendation](#).
- [Project Info](#).

PROJECTS

End-to-End Transformer Attention Inference: Cross-Layer CUDA Performance Study (Custom CUDA vs Triton vs CUTLASS vs cuBLAS)

May 2026 – Present

CUDA, Triton, CUTLASS, cuBLAS

- Optimized transformer attention kernels across CUDA, Triton, CUTLASS, and cuBLAS, achieving 3.2x lower latency and 2.7x higher memory throughput through shared-memory tiling, vectorized loads, and warp-level execution tuning.

Minimal Transformer Inference Engine with KV-Cache, Continuous Batching Scheduler, and Streaming Token Generation

Apr 2026 – Present

CUDA, PyTorch, Transformer Inference

- Built a transformer inference serving engine with KV-cache reuse and continuous batching, improving simulated multi-request throughput by 4x while reducing decode-stage latency by 35%.

Memory-Efficient Long-Context Transformer Attention using Paged KV-Cache and IO-Aware FlashAttention-Style Execution

Mar 2026 – May 2026

CUDA, FlashAttention, Long-Context Inference

- Implemented FlashAttention-style tiled attention with paged KV-cache allocation, reducing long-context memory usage by 60% and enabling scalable inference up to 128K-token contexts.

Autism Detection And Screening Assessment using ML

Feb 2025 – Apr 2025

Random Forest Classifier, ML

- Used different classifiers to evaluate performance metrics and ended up achieving 70% accuracy with Random Forest classifier among other techniques.
- Built an end-to-end ML solution enabling real-time autism screening assessment with [1.3x] model confidence threshold
- Utilize StreamLit for better user experience and interaction. [Project Info](#)

3D Graphics Engine - Ray Tracing

Jan 2023 – Apr 2023

Graphic Programming

- Used shading, anti-aliasing, super sampling, and real-time rendering for a realistic look
- Improved quality of the image processed by 37%
- Related [Blog-1](#) [Blog-2](#) [Blog-3](#)

CONTRIBUTIONS

Technical Blogger : ML & GPU Optimization

Sep 2024 – Present

- Authored 10+ technical articles on [Medium](#) covering high performance computing architectures, GPU Optimization techniques including CUDA programming patterns and practical ML problem solving methodologies with code implementations.

De-Lit - (The Literature Club)

Sep 2023 – Apr 2026

Co-founder and Member

- Started a literature club with 50+ active members, building a like-minded community for open and engaging discussions.
- Coordinated interactive sessions on deeper author perspectives on english literature and critical thinking.

CERTIFICATIONS

- Completed Coursera Machine Learning Specialization by DeepLearning.ai and Stanford Online.
- Top 5% in Industrial IOT(industrial internet of things IIOT) by NPTEL 2025.
- Gained practical Knowledge on regression, classification, neural networks, clustering, and recommendation systems. [certification](#)
- AI Fundamentals by IBM Skill Build.
- Intro to Programming by Kaggle.
- Check all my certifications [here](#)